Seunguk Lee

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Research Interests:
Computer Vision, 6D Pose Estimation,
Foundation Models, Anomaly Detection,
Smart Manufacturing

Education

2020–Present | **Sungkyunkwan University**
B.S. Candidate in System Management Engineering
Expected Graduation: Feb. 2028

Research Interests

Industrial AI for Smart Manufacturing
Computer Vision and Visual Inspection
Anomaly Detection and Predictive Quality
Foundation Models for Physical AI
6D Object Pose Estimation

Selected Coursework

Probability and Statistics,
Operations Research,
Data Science Programming,
Experimental Design,
Smart Factory/Manufacturing Engineering

Technical Skills

- Python, SQL
- PyTorch, OpenCV, NumPy, Pandas, Scikit-Learn
- Git, VS Code, Overleaf
- Statistical Analysis, Optimization, ML

Academic Projects

2026 | **FoundPose Reproduction Study**

Project Topic: Reproduction and Analysis of Training-Free 6D Pose Estimation for Unseen Objects

- Studied *FoundPose*, an RGB-only model-based method for unseen object 6D pose estimation using 3D object models and frozen DINOv2 features.
- Reproduced the core pipeline including offline template rendering, DINOv2 patch descriptor extraction, bag-of-words template retrieval, and PnP-RANSAC-based coarse pose estimation.
- Analyzed the role of intermediate DINOv2 descriptors and featuremetric refinement for handling occlusion, symmetry, and texture-less industrial objects.
- Evaluated the reproduced pipeline on the LM-O dataset by comparing local results with the reported BOP benchmark performance.

Expected Research Outcomes

Development of a reproducible training-free 6D pose estimation pipeline for unseen industrial objects
Analysis of foundation features in template-based pose estimation under occlusion, symmetry, and low-texture conditions
Evaluation of the applicability of vision-based foundation models to smart manufacturing and digital twin environments
Establishment of an experimental basis for future undergraduate research in industrial AI and vision-based automation